# SECTION 1
# ARCHITECTURE AND PASS CREATION

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## 1.1 PROJECT AIM AND SYSTEM MAP.

Manav built Private Pass, a prototype for offline face checks, to reduce ticket sharing and central face storage (figure 1).

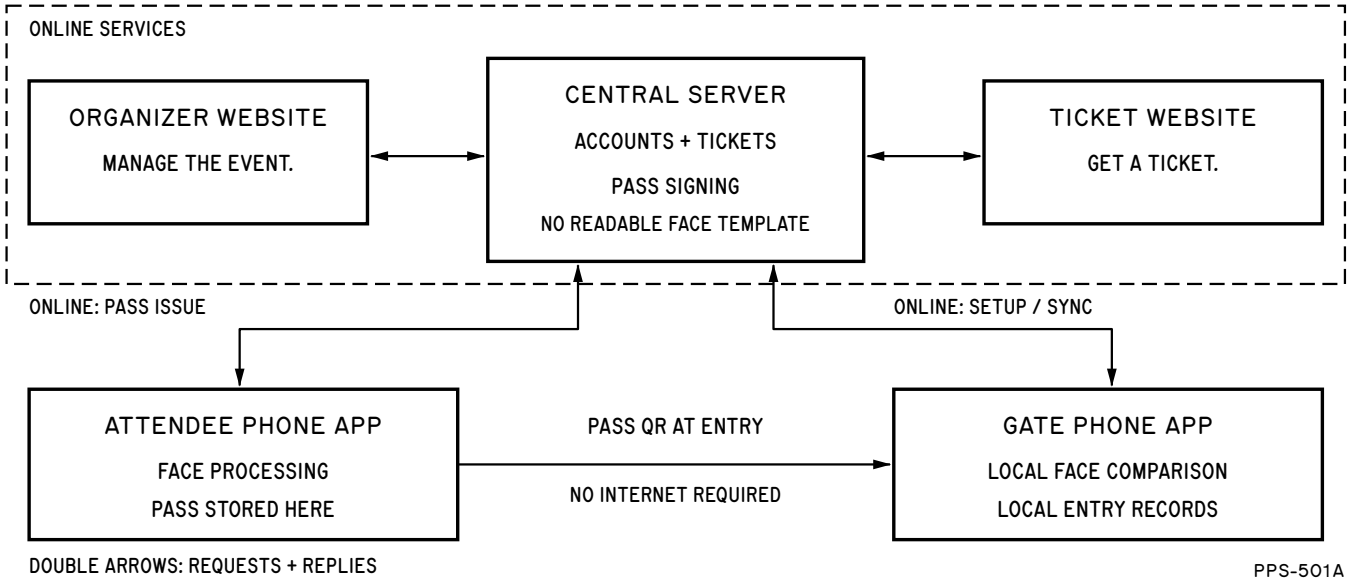



Figure 1. System Architecture: Online Links and Local Entry

PPS

## 1.2 HOW DOES A FACE BECOME A PASS?

The phone changes a face image into numbers. An event rule makes a template from those numbers. The phone encrypts it with the gate's public key. The server examines ticket ownership and signs the pass. The phone shows the signed pass as a QR code (figure 2).

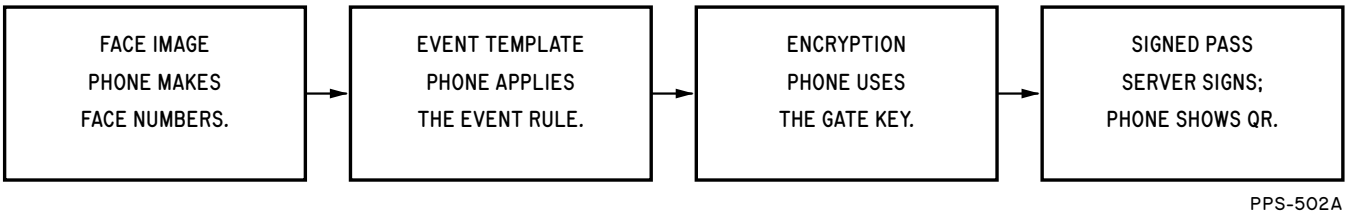



Figure 2. Pass Creation: The Face Stays on the Phone

Table 1. Essential Terms

| Term | Meaning |
|---|---|
| Pass | A digital ticket with encrypted face data. |
| Template | Numbers used for the face comparison. |
| Key | Data to lock, open, or verify. |
| Signature | Data for a source and change check. |
| Encryption | A change that limits who can read data. |
| Offline | Without an internet connection. |

**NOTE**

The server receives encrypted data, not the face image or readable template. The public key can be shared. The private key stays on the gate and opens the template. Face data still has privacy risks.